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Mannitol (systemic): Drug information

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For additional information see "[Mannitol \(systemic\): Patient drug information](#)" and "[Mannitol \(systemic\): Pediatric drug information](#)"

For abbreviations, symbols, and age group definitions [show table](#)

Brand Names: US

Osmitrol

Brand Names: Canada

Osmitrol

Pharmacologic Category

Diuretic, Osmotic

Dosing: Adult

[Collapse All](#)

Ciguatera fish poisoning

Ciguatera fish poisoning (off-label use):

Note: Patients must be fluid repleted prior to administration; do not administer to patients with significant dehydration or signs of shock ([Ref](#)).

IV (using 20% solution): 1 g/kg once administered over 30 to 45 minutes (up to 4 hours) preferably within 48 to 72 hours of consumption of contaminated fish; may repeat dose as needed for persistent or recurrent symptoms ([Ref](#)).

Intracranial pressure reduction

Intracranial pressure reduction (eg, due to cerebral edema, trauma, intracerebral hemorrhage, acute ischemic stroke, hepatic encephalopathy, transtentorial herniation syndrome) (off-label dosing):

IV (using 20% solution): 0.5 to 2 g/kg once; may repeat 0.25 to 1 g/kg per dose every 4 to 6 hours based on response and clinical status ([Ref](#)).

Intraocular pressure reduction

Intraocular pressure reduction:

Presurgical dosing: **IV:** 1.5 to 2 g/kg administered over 30 to 60 minutes 1 to 1.5 hours prior to surgery.

Traumatic hyphema: **IV:** 1.5 g/kg administered over 45 minutes twice daily for IOP >35 mm Hg; may administer every 8 hours in patients with extremely high pressure ([Ref](#)).

Kidney transplant, intraoperative volume optimization

Kidney transplant, intraoperative volume optimization (off-label use): Note:

Concentrated mannitol (ie, 20%) is preferred to optimize intravascular volume status ([Ref](#)).

IV: 12.5 to 25 g at kidney revascularization ([Ref](#)); doses up to 50 g have been studied ([Ref](#)). Some experts utilize 1 g/kg (maximum dose: 75 g); however, many centers utilize fixed dosing ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Adult

Contraindicated in severe kidney impairment. Use caution in patients with underlying kidney disease. May be used to reduce the incidence of acute kidney injury when administered prior to revascularization during kidney transplantation.

Dosing: Liver Impairment: Adult

No dosage adjustment necessary.

Dosing: Older Adult

Refer to adult dosing. Consider initiation at lower end of dosing range.

Dosing: Pediatric

(For additional information see "[Mannitol \(systemic\): Pediatric drug information](#)")

Dosage guidance:

Dosage form information: Commercially available products: 10% IV product contains 0.1 g/mL of mannitol; 20% IV product contains 0.2 g/mL of mannitol; 25% IV product contains 0.25 g/mL of mannitol.

[Collapse All](#)

Intracranial pressure reduction



Intracranial pressure (ICP) reduction: Infants, Children, and Adolescents: IV: Usual range: 0.25 to 1 g/kg/dose infused over 20 to 30 minutes; repeat as needed to maintain serum osmolality <320 mOsm/kg ([Ref](#)).

Intraocular pressure reduction



Intraocular pressure (IOP) reduction: Infants, Children, and Adolescents: IV: 1.5 to 2 g/kg/dose infused over ≥ 30 minutes. **Note:** When used preoperatively, administer 60 to 90 minutes prior to surgery.

Traumatic hyphema, intraocular pressure reduction



Traumatic hyphema, intraocular pressure (IOP) reduction: Infants, Children, and Adolescents: IV: 1.5 g/kg/dose infused over 45 minutes twice daily for IOP >35 mm Hg; may administer every 8 hours in patients with extremely high pressure ([Ref](#)).

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling for kidney impairment. Use with caution in patients with underlying renal disease, with conditions that put them at high risk for developing kidney failure, or receiving potentially nephrotoxic drugs. Use is contraindicated in patients with anuria.

Dosing: Liver Impairment: Pediatric

There are no dosage adjustments provided in the manufacturer's labeling.

Adverse Reactions

The following adverse drug reactions are derived from product labeling unless otherwise specified.

Frequency not defined:

Cardiovascular: Chest pain, edema, heart failure, hypotension, palpitations, peripheral edema, tachycardia, thrombophlebitis

Dermatologic: Skin necrosis, urticaria

Endocrine & metabolic: Dehydration, hyperkalemia, hypernatremia, hypervolemia, hypokalemia, hyponatremia, hypovolemia, increased thirst, metabolic acidosis, metabolic alkalosis

Gastrointestinal: Nausea, vomiting, xerostomia

Genitourinary: Anuria, azotemia, diuresis, hematuria, oliguria, osmotic nephrosis, polyuria, urinary retention

Hematologic & oncologic: Hemoconcentration

Hypersensitivity: Hypersensitivity reaction (including anaphylaxis, severe hypersensitivity reaction)

Local: Infusion-site reaction (including local inflammation, local pain, local pruritus, localized erythema, localized phlebitis, localized rash)

Nervous system: Asthenia, chills, coma, confusion, dizziness, headache, increased intracranial pressure (rebound), lethargy, malaise, seizure

Neuromuscular & skeletal: Arm and/or wrist pain

Ophthalmic: Blurred vision

Renal: Acute kidney injury

Respiratory: Pulmonary congestion, pulmonary edema, rhinitis

Miscellaneous: Fever

Contraindications

Hypersensitivity to mannitol or any component of the formulation; anuria; severe hypovolemia; active intracranial bleeding except during craniotomy; preexisting severe pulmonary vascular congestion or pulmonary edema.

Warnings/Precautions

Concerns related to adverse effects:

- Extravasation: Vesicant (at concentrations >5%); ensure proper catheter or needle position prior to and during IV infusion. Avoid extravasation of IV infusions; may

cause compartment syndrome. Administration into a large central vein is recommended.

- **Fluid/electrolyte imbalance:** May cause hypervolemia and electrolyte disturbances; monitor for new onset or worsening cardiac or pulmonary congestion. Also may cause profound diuresis with fluid and electrolyte loss; close medical supervision and dose evaluation are required. Correct electrolyte disturbances; adjust dose to avoid dehydration.
- **Hypersensitivity:** Serious hypersensitivity reactions (eg, anaphylaxis), including fatalities, have been reported. Discontinue mannitol immediately if hypersensitivity reaction develops and treat accordingly.
- **Nephrotoxicity:** May cause kidney dysfunction, especially with high doses; use caution in patients taking other nephrotoxic agents, with sepsis, or preexisting kidney disease. To minimize adverse kidney effects, monitor serum osmolality or osmolar gap (NCCS [Cook 2020]). Discontinue mannitol if acute kidney injury develops.

Disease-related concerns:

- **Administration:** Do not administer mannitol solutions simultaneously with blood due to the possibility of pseudoagglutination or hemolysis.
- **Cardiac and pulmonary:** Mannitol may initially worsen pulmonary edema or heart failure by causing hypervolemia before diuresis ensues. Do not administer large volumes of mannitol to patients with preexisting severe pulmonary edema or heart failure. Discontinue mannitol if cardiac or pulmonary status worsens.
- **Cerebral edema:** In patients being treated for cerebral edema, mannitol may accumulate in the brain (causing rebound increases in intracranial pressure) if circulating for long periods of time as with continuous infusion; intermittent boluses preferred. If hypotension occurs, monitor cerebral perfusion pressure; reassess dose and mannitol therapy if there is a decrease in cerebral perfusion pressure.
- **CNS effects:** CNS toxicity (eg, coma, confusion, lethargy) may occur; risk may be increased in patients with impaired kidney function or with concomitant use of neurotoxic drugs. Discontinue mannitol if CNS toxicity develops.
- **Kidney impairment:** Use with caution. In patients with severe impairment; do not use until adequacy of kidney function and urine flow is established.

Warnings: Additional Pediatric Considerations

Mannitol may increase cerebral blood flow, increase the risk of postoperative bleeding in neurosurgical patients, and worsen intracranial hypertension in children who develop

generalized cerebral hyperemia during the first 24 to 48 hours after traumatic brain injury.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling. [DSC] = Discontinued product

Solution, Intravenous:

Osmitrol: 10% (500 mL); 15% (500 mL [DSC]); 20% (250 mL, 500 mL)

Generic: 20% (500 mL); 25% (50 mL)

Solution, Intravenous [preservative free]:

Generic: 20% (250 mL, 500 mL)

Generic Equivalent Available: US

Yes

Pricing: US

Solution (Mannitol Intravenous)

20% (per mL): \$0.05 - \$0.12

25% (per mL): \$0.15 - \$0.22

Solution (Osmitrol Intravenous)

10% (per mL): \$0.12

20% (per mL): \$0.26

Disclaimer: A representative AWP (Average Wholesale Price) price or price range is provided as reference price only. A range is provided when more than one manufacturer's AWP price is available and uses the low and high price reported by the manufacturers to determine the range. The pricing data should be used for benchmarking purposes only, and as such should not be used alone to set or adjudicate any prices for reimbursement or purchasing functions or considered to be an exact price for a single product and/or manufacturer. Medi-Span expressly disclaims all warranties of any kind or nature, whether express or implied, and assumes no liability with respect to accuracy of price or price range data published in its solutions. In no event shall Medi-Span be liable for special, indirect, incidental, or consequential damages arising from use of price or price range data. Pricing data is updated monthly.

Dosage Forms: Canada

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Intravenous:

Osmitrol: 10% (1000 mL); 20% (500 mL)

Generic: 20% (500 mL); 25% (50 mL)

Administration: Adult

IV: Concentration and rate of administration depends on indication/severity. Use of an administration set with a final in-line filter (eg, ≤ 5 micron) is recommended ([Ref](#)). For cerebral edema or elevated intracranial pressure, administer over 10 to 30 minutes; for patients with high risk for cerebral herniation, use faster administration rates (over ~ 10 minutes) ([Ref](#)). Inspect for crystals prior to administration. If crystals are present, redissolve by warming solution. Do not place mannitol 25% injection in PVC bags; a white flocculent precipitate may form from contact with PVC surfaces. Do not administer simultaneously with blood due to the possibility of pseudoagglutination or hemolysis. If coadministration with blood is essential, at least 20 mEq sodium chloride should be added to each liter of mannitol.

Vesicant (at concentrations $>5\%$); ensure proper catheter or needle position prior to and during IV infusion. Avoid extravasation of IV infusions. Administration into a large central vein is recommended.

Extravasation management: If extravasation occurs, stop infusion immediately; leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution then remove needle/cannula; elevate extremity; apply dry warm compresses; initiate hyaluronidase antidote ([Ref](#)).

Hyaluronidase: **Intradermal or SUBQ:** Inject a total of 1 mL (15 units/mL) as 5 separate 0.2 mL injections (using a tuberculin syringe) around the site of extravasation; if IV catheter remains in place, administer IV through the infiltrated catheter; may repeat in 30 to 60 minutes if no resolution ([Ref](#)).

Administration: Pediatric

Note: Commercially available products: 10% IV product contains 0.1 g/mL of mannitol; 20% IV product contains 0.2 g/mL of mannitol; 25% IV product contains 0.25 g/mL of mannitol.

Parenteral: Administer IV only; do not administer IM or SUBQ. Central line preferred. Use of an inline filter set (≤ 5 microns) is recommended ([Ref](#)). Inspect for crystals prior to administration; if crystals are present, redissolve by warming solution; see manufacturer

labeling for details. Inspect for particulates or discoloration; if particulates or discoloration, discard bag. Do not place mannitol 25% injection in PVC bags; a white flocculent precipitate may form from contact with PVC surfaces. Do not administer with blood; pseudoagglutination or hemolysis of red blood cells may occur. Infusion rate is dependent upon indication:

Cerebral edema or increased intracranial pressure: Administer over 20 to 30 minutes ([Ref](#)).

Increased intraocular pressure: Administer over ≥ 30 minutes.

Traumatic hyphema: Administer over 45 minutes ([Ref](#)).

Vesicant (at concentrations $>5\%$); ensure proper catheter or needle position prior to and during IV infusion. Avoid extravasation of IV infusions. If extravasation occurs, stop infusion immediately; leave needle/cannula in place temporarily but do **NOT** flush the line; gently aspirate extravasated solution, then remove needle/cannula unless needed for IV antidote; elevate extremity; apply dry, warm compresses; initiate hyaluronidase antidote ([Ref](#)).

Vesicant/Extravasation Risk

Vesicant (at concentrations $>5\%$)

Storage/Stability

Store at controlled room temperature; do not freeze. Crystallization of solution may occur at low temperatures; do not use solutions that contain crystals.

To dissolve crystals in plastic containers, heat solution up to 60°C to 70°C (140°F to 158°F) (recommended temperature varies by product; refer to manufacturer's labeling), with agitation; do not immerse in water or microwave (may contaminate or damage product); allow solution to return to room or body temperature before use.

To dissolve crystals in vials/bottles, warm vial in hot water at 80°C (176°F) and periodically shake vigorously, or vial may be autoclaved at 121°C (250°F) for 20 minutes at 15 psi. Allow solution to return to room or body temperature before use.

Use: Labeled Indications

Injection: Reduction of increased intracranial pressure (associated with cerebral edema and/or brain mass); reduction of increased intraocular pressure.

Use: Off-Label: Adult

Ciguatera fish poisoning; Kidney transplant, intraoperative volume optimization

Medication Safety Issues

Sound-alike/look-alike issues:

Osmitol may be confused with esmolol

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: [Launch drug interactions program](#))

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within “CYP3A4 Inducers [Strong]” are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the “Launch drug interactions program” link above.

Amikacin Liposome (Oral Inhalation): May increase nephrotoxic effects of Mannitol (Systemic).

Risk X: Avoid

Aminoglycosides: Mannitol (Systemic) may increase nephrotoxic effects of Aminoglycosides.

Management: Consider avoiding coadministration of aminoglycosides and mannitol. If combined, consider monitor for increased aminoglycoside toxicity (eg, nephrotoxicity, ototoxicity). *Risk D: Consider Therapy Modification*

Arsenic Trioxide: Osmotic Diuretics may increase QTc-prolonging effects of Arsenic Trioxide.

Management: When possible, avoid concurrent use of arsenic trioxide with drugs that can cause electrolyte abnormalities, such as the osmotic diuretics. *Risk D: Consider Therapy Modification*

Baxdrostat: Hyponatremia-Associated Agents may increase hyponatremic effects of Baxdrostat.

Risk C: Monitor

Desmopressin: Hyponatremia-Associated Agents may increase hyponatremic effects of Desmopressin.

Risk C: Monitor

Diacerein: May increase therapeutic effects of Diuretics. Specifically, the risk for dehydration or hypokalemia may be increased. *Risk C: Monitor*

Isocarboxazid: May increase hypotensive effects of Diuretics. *Risk X: Avoid*

Lithium: Osmotic Diuretics may decrease serum concentrations of Lithium. Management: Consider temporarily discontinuing use of osmotic diuretics during coadministration with lithium. If coadministration is required, monitor lithium concentrations and for signs of lithium toxicity frequently. *Risk D: Consider Therapy Modification*

Mivacurium: Mannitol (Systemic) may increase therapeutic effects of Mivacurium. *Risk C: Monitor*

Opioid Agonists: May increase adverse/toxic effects of Diuretics. Opioid Agonists may decrease therapeutic effects of Diuretics. *Risk C: Monitor*

Piperacillin: May increase hypokalemic effects of Diuretics. *Risk C: Monitor*

Polyethylene Glycol-Electrolyte Solution: Diuretics may increase nephrotoxic effects of Polyethylene Glycol-Electrolyte Solution. *Risk C: Monitor*

Sodium Phosphates: Diuretics may increase nephrotoxic effects of Sodium Phosphates. Specifically, the risk of acute phosphate nephropathy may be enhanced. *Risk C: Monitor*

Tobramycin (Oral Inhalation): Mannitol (Systemic) may increase adverse/toxic effects of Tobramycin (Oral Inhalation). Specifically, the risk for nephrotoxicity and ototoxicity may be increased. *Risk X: Avoid*

Pregnancy Considerations

Mannitol crosses the placenta.

Outcome information following surgical use in pregnancy is limited; amniotic fluid volume may be decreased (Handlogten 2015; Kazemi 2014).

Breastfeeding Considerations

It is not known if mannitol is present in breast milk.

According to the manufacturer, the decision to breastfeed during therapy should consider the risk of infant exposure, the benefits of breastfeeding to the infant, and benefits of treatment to the mother. Other sources consider mannitol to be compatible with breastfeeding (WHO 2002).

Monitoring Parameters

Kidney function; cardiac and pulmonary function; urine output; intravascular volume status; serum electrolytes, serum osmolality (measured); osmolar gap; acid-base status; serum glucose; intracranial pressure (if applicable). Monitor infusion site.

For treatment of cerebral edema and intracranial pressure, some suggest monitoring osmolar gap as a surrogate marker for mannitol accumulation to evaluate for the risk of acute kidney injury (NCCS [Cook 2020]). Others suggest maintaining serum osmolality <320 mOsm/kg due to the potential risk of acute kidney injury; however, this threshold may be exceeded if clinically necessary (Grape 2012; Rabenstein 2006).

Mechanism of Action

Produces an osmotic diuresis by increasing the osmotic pressure of glomerular filtrate, which inhibits tubular reabsorption of water and electrolytes and increases urinary output. Mechanism of action in reduction of intracranial pressure (ICP) is less clear. However, it is thought that mannitol reduces ICP by reducing blood viscosity which transiently increases cerebral blood flow and oxygen transport and constricts pial arterioles. This in turn reduces cerebral blood volume and ICP. Furthermore, mannitol reduces ICP by withdrawing water from the brain parenchyma and excretes water in the urine (Allen 1998; BTF [Carney 2017]).

For the treatment of ciguatera fish poisoning, mannitol appears to reduce neuronal edema and has been found to reverse ciguatoxin-induced neurotoxic effects by reducing sodium entry and increasing the efflux of water (Benoit 1996).

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: Diuresis: 1 to 3 hours; Reduction in intracranial pressure: ~15 to 30 minutes.

Duration: Reduction in intracranial pressure: 1.5 to 6 hours.

Distribution: 17 L; remains confined to extracellular space (except in extreme concentrations); does not penetrate the blood-brain barrier (generally, penetration is low).

Metabolism: Minimally hepatic to glycogen.

Half-life elimination: 0.5 to 2.5 hours; ~36 hours in acute kidney injury and end-stage kidney failure.

Excretion: Urine (~80% as unchanged drug).

Pharmacokinetics: Additional Considerations (Adult Data Unless Noted)

Altered kidney function: Elimination half-life is ~36 hours in patients with acute kidney injury and end-stage kidney failure. In patients with kidney impairment receiving hemodialysis, half-life elimination reduced to ~6 hours; in patients receiving peritoneal dialysis, half-life elimination reduced to ~21 hours.

Patient Counseling Points

(For additional information see "[Mannitol \(systemic\): Patient drug information](#)")

What is this drug used for?

- It is used to lower pressure in the skull and brain.
- It is used to lower high eye pressure.
- It may be given to you for other reasons. Talk with the doctor.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

- Headache
- Upset stomach or throwing up
- Feeling dizzy, tired, or weak
- Sweating a lot

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

- Kidney problems like unable to pass urine, change in how much urine is passed, blood in the urine, or a big weight gain
- Fluid and electrolyte problems like mood changes, confusion, muscle pain or weakness, fast or abnormal heartbeat, severe dizziness or passing out, increased thirst, seizures, feeling very tired or weak, decreased appetite, unable to pass urine or change in the amount of urine produced, dry mouth, dry eyes, or severe upset stomach or throwing up.
- Too much acid in the blood (acidosis) like confusion; fast breathing; fast heartbeat; a heartbeat that does not feel normal; very bad stomach pain, upset stomach, or throwing up; feeling very sleepy; shortness of breath; or feeling very tired or weak
- High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight
- Pain and irritation where this drug goes into the body
- Change in skin color to black or purple
- Skin breakdown where this drug is used

- Shortness of breath, a big weight gain, or swelling in the arms or legs
- A heartbeat that does not feel normal
- Feeling confused
- Feeling sluggish
- Blurred eyesight
- Fever or chills
- Muscle stiffness
- Muscle pain
- Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country



For country code abbreviations ([show table](#))

(AE) United Arab Emirates: Osmitol; (AR) Argentina: Manitol norgreen | Solucion d manitol | Solucion de manitol | Solucion de manitol al 15% en agua | Solucion de manitol al 15% rigecin; (AT)

Austria: Mannit; (AU) Australia: Mannitol/osmitrol; (BD) Bangladesh: Manisol | Mannisol | Menitol | Osmosal; (BE) Belgium: Mannitol 15% baxter; (BR) Brazil: Endonitol | Hypofarma manitol | Manitol | Manitol baxter | Solucao de manitol | Solucao de manitol equiplex; (CH) Switzerland: Mannitol braun; (CN) China: Pu ke; (CO) Colombia: Manitol | Osmorin Manitol | Osmotrol; (CR) Costa Rica: Manitol | Osmorol; (CZ) Czech Republic: Ardeaosmosol | Ardeaosmosol ma | Infusio mannitoli | Manit | Manitol viaflo | Osmofundin n; (DE) Germany: Deltamannit | Mannit | Thomaemannit; (DO) Dominican Republic: Acrosmosol | Manitol | Osmorol; (EC) Ecuador: Manitol | Osmorin; (EE) Estonia: Mannit; (EG) Egypt: Mannitol water; (ES) Spain: Apiros manitol | Manitol baxter | Manitol mein | Manitul vitulia | Osmofundina concentrada | Suero manitol vitulia; (FI) Finland: Mannitol baxter | Mannitol braun | Mannitol Fresenius Kabi | Mannitol medipolar; (FR) France: Mannitol aguetant | Mannitol b.braun | Mannitol baxter | Mannitol bieffe | Mannitol cooper | Mannitol fresenius | Mannitol lavoisier | Mannitol macopharm; (GB) United Kingdom: Flowfusor mannitol | Intravenous mannitol | Osmitol | Polyfusor mannitol | Trav mannitol(osm); (GR) Greece: Mannitol/bradex | Mannitol/fresenius; (HK) Hong Kong: Mannitol in water; (HR) Croatia: Manit | Manitol HZTM; (HU) Hungary: Mannisol | Mannitol bieffe | Mannitoli Imuna | Mannozym; (ID) Indonesia: Infusan M | Manitol; (IN) India: P tol; (IT) Italy: Isotol | Mannistol | Mannitolo | Sol Mannitolo; (JP) Japan: D mannitol fuso | Manitol taiho | Maniton s | Mannigen | Mannit t | Mannite kyowa hakko | Mannitol daiichi | Mannitol kawasumi | Mannitol kobayashi | Mannitol Kowa | Mannitol kyosei | Mannitol nichiiiko | Mannitol nikken | Mannyl; (KE) Kenya: Infusol M10 | Infusol M20 | Osmosol | Osmosteril; (KR) Korea, Republic of: Daihan d mannitol | Inno.n 15% mannitol | Inno.n 20% mannitol | Jw 15% mannitol | Jw 20% mannitol | Jw 25% mannitol; (LT) Lithuania: Mannisol a | Mannisol b | Mannit; (LV) Latvia: Manit | Mannit; (MX) Mexico: Manitol | Osmokab | Osmorol; (MY) Malaysia: Infusol M10 | Infusol M20; (NG) Nigeria: Unimann; (NL) Netherlands: Osmosteril; (NO) Norway: Mannitol baxter viaflo; (NZ) New Zealand: Osmitol c water; (PA) Panama: Manitol; (PE) Peru: Demanitol | Diurecide | Manitol; (PH) Philippines: Osmitol; (PK) Pakistan: Alimanit | Manitol | Mannitozaf | Medisol mannitol | Osmofundin | Osmotol | Revasol m; (PT) Portugal: Manitol; (PY) Paraguay: D manitol sanderson | Manitol demotek | Manitol icu vita | Solucion de d manitol roux ocefa | Solucion de manitol b.braun | Solucion de manitol fusa; (RO) Romania: Osmofundin; (RU) Russian Federation: Mannit; (SE) Sweden: Mannitol baxter viaflo | Mannitol Fresenius Kabi; (SI) Slovenia: Manit; (SK) Slovakia: Mannitol Viaflo | Osmofundin; (SR) Suriname: Manitol; (SV) El Salvador: Osmorol; (TH) Thailand: Maniton | Mannitol 20% a.n.b | Mannitol thai otsu; (TR) Turkey: Mannitol mediflex | Neofleks Mannitol | Polifleks mannitol | Rezosel; (TW) Taiwan: Anol | Maniton | Sunnytol; (UA) Ukraine: Mannit; (UG) Uganda: Abtol | Mannilife; (UY) Uruguay: Manitol | Solucion de manitol farmaco uruguayo; (VE) Venezuela, Bolivarian Republic of: Manitol; (ZM) Zambia: 20 m | Intramed mannitol | Sabax osmitrol

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